

EMCH 290
Mechanical Engineering Thermodynamics

Homework # 1

Points Possible: 20

Problem No.1

Find the change in gravitational potential energy of the drone ΔPE , in ft·lbf and BTU?

(1 ft·lbf = 0.0013 BTU).

(5 points)

Problem No. 2

An object whose weight is 100 lbf experiences a decrease in kinetic energy of 500 ft·lbf and an increase in potential energy of 1500 ft·lbf. The initial velocity and elevation of the object, each relative to the surface of the earth, are 40 ft/s and 30 ft, respectively. If $g=32.2 \text{ ft/s}^2$, determine:

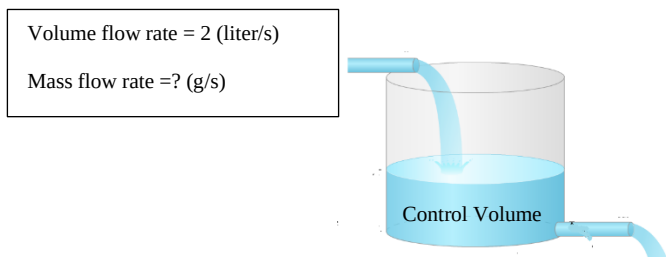
(a) The final velocity, in ft/s.

(b) The final elevation, in ft.

(5 points)

Problem No. 3

Find the mass flow rate (in gram per second) of water for a control volume system, if the volume flow rate is 2 liter per second. (5 points)



Problem No. 4

What is the change in the internal energy of the system if the surrounding releases 300 J of heat energy and if the system does 550 J of work on the surroundings?

(5 points)

Formula:

$$\Delta U = Q + w$$

Q = + endo

W = + on the system

Q = - exo

W = - by the system